

Tropical Cyclone Genesis II

Learning goals. By the end of class, you should be able to:

- describe the conceptual idea behind convective self-aggregation;
- explain, in words, the instability that causes moist and dry regions to separate;
- explain how rotation connects self-aggregation to tropical cyclogenesis; and
- describe the Marsupial Pouch Theory for tropical cyclone genesis.

1. Central Question

Central question: Can tropical cyclogenesis be viewed partly as _____?

Complete the conceptual pathway discussed in class:



Important caveat: The lecture does *not* argue that all tropical cyclones form by self-aggregation. In one or two sentences, write what the self-aggregation framework is being used for here.

2. Convective Self-Aggregation

Definition. In your own words, define convective self-aggregation:

In idealized radiative–convective equilibrium (RCE), convection can cluster even when the following are horizontally homogeneous:

- sea-surface temperature: _____
- large-scale forcing: _____
- boundary conditions: _____

What does the aggregated state look like?

Moist region

Dry region

What variable(s) can be used to diagnose the organization?

3. Moist Static Energy (MSE)

The lecture defines moist static energy as

$$h = c_p T + gz + L_v q$$

Fill in the physical meaning of each term:

$$c_p T$$

$$gz$$

$$L_v q$$

Why is MSE useful for tropical convection?

For approximately moist-adiabatic motion,

$$h \approx \text{_____}.$$

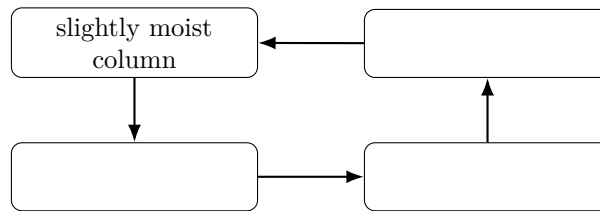
This means MSE helps us follow exchanges among

$$\text{_____} \longleftrightarrow \text{_____} \longleftrightarrow \text{_____}.$$

Physical interpretation near the surface: High MSE generally means the air is _____ and _____, making it favorable for _____.

4. The Instability Behind Self-Aggregation

Start with a small perturbation: Suppose one column becomes slightly moister than its surroundings. Complete the positive feedback loop:



Key idea: The instability increases the spatial variance of MSE. Translate this into words:

moist columns become _____ and dry columns become _____.

Why is this called “upgradient” transport?

5. MSE Variance Budget: What Counts as a Positive Feedback?

The lecture uses the schematic budget

$$\frac{1}{2} \frac{d}{dt} \left(\widehat{\text{MSE}}' \right)^2 = \left\langle \widehat{\text{MSE}}' F'_{sfc} \right\rangle + \left\langle \widehat{\text{MSE}}' Q'_{rad} \right\rangle - \left\langle \widehat{\text{MSE}}' C'_h \right\rangle.$$

You do *not* need to memorize this equation. Use it to answer the conceptual questions below.

1. What does a positive contribution to this budget mean physically?
2. Why is “more heating” by itself *not* enough to define a positive feedback?
3. What matters about the spatial relationship between heating/cooling and MSE anomalies?

6. Why Radiation Matters

Mechanism-denial experiment: What happens when radiative cooling is horizontally homogenized?

Longwave-radiation mechanism:

Dry region	Moist/cloudy region
LW cooling: _____	LW cooling: _____
Vertical motion: _____	Vertical motion: _____

How does this differential radiative forcing generate a shallow circulation that favors aggregation?

7. Other Feedbacks

For each process, write one sentence explaining how it can influence convective organization.

1. **Entrainment / environmental humidity:**
2. **Surface enthalpy fluxes:**
3. **Cold pools:**
4. **Boundary-layer gravity waves:**

8. Add Rotation: From Aggregation to Cyclogenesis

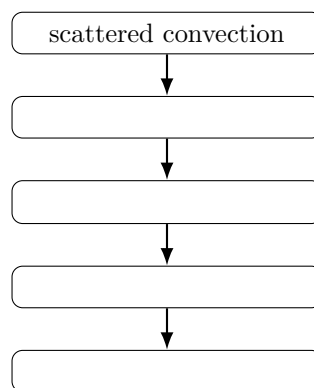
Key transition: In rotating RCE, an aggregated moist convective region can acquire _____

From the Muller and Romps (2018) experiments, complete the interpretation:

Experiment	What happens?
Control	
No radiative feedbacks	
No radiative + no surface-flux feedbacks	

Interpretation: Why can self-aggregation feedbacks accelerate tropical cyclogenesis?

Complete the sequence:



9. Limits of the Self-Aggregation View

Write one limitation or caveat for each category.

Idealized setup

Timescale

Real-world seed disturbances

Multiple mechanisms

Bottom line: In nature, self-aggregation is best viewed as _____

10. Marsupial Paradigm: The Basic Idea

Conceptual shift: Tropical cyclogenesis occurs inside a _____ embedded in a tropical wave.

The favored location is the **sweet spot**: the intersection of

1. the _____, and
2. the _____.

At the critical latitude,

$$\bar{u}(y_c) \approx \underline{\hspace{2cm}},$$

where the right-hand side represents the _____.

Sketch the pouch. Draw a wave-relative streamline pattern and mark:

- the wave trough;
- the critical latitude;
- the “sweet spot”;
- the region where air parcels are relatively protected.

11. Why Is the Pouch a Favorable Incubator?

Complete the three ideas emphasized in the lecture.

Protection The pouch reduces _____

**Repeated moist-
ening** Air parcels remain near _____

**Vorticity gather-
ing** The _____ wave/critical-layer _____ flow _____ helps _____ concentrate

Complete the conceptual pathway:



12. Vortex Formation Inside the Pouch

Complete the sequence in words.

1. Convective bursts stretch _____.
2. Convectively generated vortices can _____.
3. Diabatic heating helps build a positive _____ anomaly.
4. Progressive moistening makes convection more _____ and _____.

Why does keeping convection, moisture, and cyclonic vorticity collocated matter?

13. From Pouch to Tropical Cyclone

The pouch does not guarantee genesis. What additional transition must occur?

Complete the final sequence:

easterly-wave pouch → _____ → _____ → self-sustaining TC vortex

14. Compare the Two Frameworks

Complete the table during the synthesis portion of the lecture.

	Marsupial pouch	Convective self-aggregation
Starting point		
Organizer		
Protected / favored region		
Role of moisture		
Role of vorticity		
Best mental model		

Common thread: What do both frameworks suggest must remain collocated long enough for genesis to occur?

15. Reflection

Answer each in 1–3 sentences.

1. Why can a small moisture anomaly grow into a larger moist–dry contrast in self-aggregation?
2. What is the role of longwave radiative cooling in the instability?
3. Why does adding rotation change the significance of a moist convective aggregate?
4. What does the Marsupial Paradigm add that is missing from homogeneous self-aggregation experiments?